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Confidence Calibration and Reliability Assessment in Multi-Agent LLM Systems: A Focused Structured Review

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ABSTRACT

Large Language Models (LLMs) have demonstrated strong performance in various natural language processing tasks, but their ability to accurately estimate and communicate confidence in their outputs remains a significant challenge. High task accuracy does not necessarily imply that a model's confidence reflects the likelihood of correctness. This focused structured review examines the development of confidence estimation, uncertainty quantification, and reliability assessment from conventional neural-network calibration to modern LLM and multi-agent settings. The review first analyzes five foundational studies covering probabilistic calibration, model self-evaluation, reasoning consistency, evidence-based multi-hop reasoning, and broader LLM confidence estimation. It then incorporates recent work on multi-agent confidence calibration, trajectory-level uncertainty quantification, runtime calibration, communication-aware calibration, and epistemic calibration during multi-agent planning. The selected studies are analyzed according to their objectives, confidence or reliability signals, methodologies, evaluation strategies, and limitations. The synthesis highlights a transition from individual-model confidence toward system-level reliability and identifies five research gaps: local-to-system calibration and confidence propagation, correlated errors and agent disagreement, standardized multi-agent calibration benchmarks, explainable confidence traces, and runtime shift with computational cost. The review concludes by outlining corresponding future directions for reliable confidence calibration and system-level assessment in multi-agent LLM systems.

Keywords— Large Language Models, Confidence Estimation, Calibration, Uncertainty Quantification, Multi-Agent Systems, Self-Evaluation, Self-Consistency, Reliability Assessment

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Introduction

Large Language Models (LLMs) have shown strong abilities in many natural language processing tasks, such as answering questions, reasoning, and creating text. However, having good performance in these tasks does not always mean a model can accurately judge if its outputs are correct. This distinction is especially important for systems that use model outputs to support decisions, perform multiple steps of reasoning, or interact with other parts of a system. Therefore, confidence estimation and calibration have become key factors in assessing the reliability of language models.

Confidence calibration is based on probabilistic predictions made by neural networks. Guo et al. (**guo2017?**) showed that even when neural networks have high classification accuracy, they can still be poorly calibrated. Their work found that the confidence a model assigns to its predictions does not always match the actual chance that the prediction is correct. They also looked at methods for post-hoc calibration, and temperature scaling was found to be an effective technique. This study highlighted an important principle: accuracy alone is not enough when a model needs to express how confident it is in its predictions.

As Large Language Models become more common, confidence estimation becomes more difficult because these models generate sequences of tokens instead of choosing a single class label. Kadavath et al. (**kadavath2022?**) explored whether language models can assess how confident they are in their answers and whether they know the correct answer to a question. Their work introduced $P(\text{True})$, which represents the probability that an answer is correct, and $P(\text{IK})$, where "IK" stands for "I Know," indicating the model's belief that it knows the answer. Their research showed that language models can perform useful self-evaluation, but also found that calibration of knowledge estimates can be more complex when models face new tasks.

Another source of reliability information comes from consistency in reasoning processes. Wang et al. (**wang2022?**) introduced self-consistency as a decoding method for chain-of-thought reasoning, where multiple reasoning paths are sampled and the answer that appears most consistent across these paths is selected. While the initial goal of self-consistency was to improve reasoning accuracy, Wang et al. (**wang2022?**) also report that consistency is highly correlated with accuracy and suggest using it as an uncertainty estimate. This makes self-consistency directly relevant to confidence assessment rather than merely an accuracy-improvement technique.

Confidence signals are especially important in complex, multi-step reasoning tasks. Yang et al. (2018) (**yang2018?**) created HOTPOTQA, a dataset for multi-hop question-answering, where questions require models to find and process information from multiple sources. The dataset also includes sentence-level supporting facts, allowing evidence to be evaluated alongside the final answer. In its distractor development setting, the reported answer EM is 44.44, supporting-fact EM is 21.95, and joint EM is 11.56 (**yang2018?**). These results illustrate why final-answer correctness alone can hide intermediate evidence errors.

Recent research has increasingly treated confidence estimation and calibration as a unique challenge for LLMs. Geng et al. (**geng2024?**) categorized existing methods into white-box and black-box approaches. White-box methods use internal information like token probabilities, model states, and semantic representations, while black-box methods rely on linguistic confidence, consistency in generated responses, and surrogate models. This broader perspective shows that confidence in LLMs cannot

be captured by a single signal and may instead need to combine probabilistic, semantic, and behavioral information.

A key limitation of earlier confidence and reliability research is that much of the evaluation has focused on individual models, individual generations, or isolated reasoning processes. More recent work has begun to examine confidence directly within multi-agent systems. Yang et al. (**yang2024?**) proposed Collaborative Calibration, a post-hoc, training-free approach that uses multiple tool-augmented LLM agents in a simulated deliberation process to improve confidence calibration. More recent research has further examined uncertainty quantification and calibration specifically in multi-agent systems, including uncertainty arising from communication dynamics, reasoning trajectories, and agent coordination (**chen2026matu?**; **armstrong2026margin?**; **huang2026cagecal?**). These developments indicate that confidence in multi-agent systems cannot be understood solely as an individual-model property.

However, the emerging multi-agent literature also introduces new questions concerning confidence propagation, correlated failures, communication topology, runtime adaptation, and the relationship between local agent confidence and global system reliability (**huang2026cagecal?**; **wang2026epistemic?**). Therefore, this review connects foundational confidence-calibration research with recent multi-agent approaches to examine how confidence estimation and reliability assessment evolve when multiple LLM agents interact.

Objectives

The primary objective of this structured review is to synthesize research on confidence calibration and uncertainty quantification in Large Language Models (LLMs), examining how confidence is estimated, evaluated, and calibrated at the model and reasoning levels, and identifying the methodological challenges and open research directions involved in extending these approaches to multi-agent LLM systems.

The specific objectives of this review are as follows:

1. **Establish a taxonomy of confidence and reliability signals:** Classify the selected literature according to six confidence or reliability signals: probabilistic calibration, model self-evaluation, consistency-based signals, evidence-based evaluation, interaction- and communication-based signals, and broader confidence-estimation approaches. The review further distinguishes whether these signals are evaluated at the local, process, or system level.
2. **Examine confidence and calibration evaluation:** Analyze how the selected literature evaluates whether model confidence corresponds to actual correctness, considering calibration measures, self-evaluation behavior, agreement across multiple reasoning paths, and evidence-supported reasoning.
3. **Compare methodological strengths and limitations:** Synthesize the assumptions, strengths, limitations, computational requirements, and reported evaluation outcomes of different confidence estimation strategies across the selected literature.
4. **Investigate uncertainty in multi-step and multi-agent systems:** Examine how uncertainty can arise and propagate across reasoning steps, agent interactions, information exchange, and intermediate decisions, and assess whether individual-agent confidence is sufficient to characterize the reliability of an overall multi-agent LLM system.
5. **Identify research gaps and future directions:** Identify unresolved challenges in multi-agent

confidence calibration and uncertainty quantification, including standardized evaluation protocols, uncertainty propagation, correlated errors between agents, interpretability of confidence estimates, computational efficiency, and the relationship between local agent confidence and overall system reliability.

Review Methodology

This review uses a structured approach to examine confidence calibration and uncertainty quantification in Large Language Models (LLMs), with a specific focus on the challenges that arise when confidence estimation is applied to multi-agent LLM systems. The review is organized around clearly defined research questions, study selection criteria, a categorization framework, and a structured approach to comparing and summarizing findings.

The methodology connects work on neural-network calibration with more recent approaches involving LLM self-evaluation, reasoning consistency, evidence-based reasoning, and broader confidence-estimation frameworks. Rather than treating all selected studies as direct calibration methods, the review distinguishes between approaches that explicitly estimate or calibrate confidence and studies that provide supporting evidence for reliability assessment.

3.1 Research Questions

The review is guided by the following research questions:

1. **RQ1:** How is confidence calibration defined and measured in machine learning and Large Language Models?
2. **RQ2:** How well can language models estimate the likelihood that their answers are correct, and how does LLM self-evaluation differ from traditional probabilistic calibration?
3. **RQ3:** Can agreement among multiple generated reasoning paths serve as a useful indicator of reasoning reliability or uncertainty?
4. **RQ4:** How should confidence and reliability be evaluated in tasks involving multiple steps or hops, where errors may come from intermediate reasoning or evidence selection, not just the final answer?
5. **RQ5:** What categories of confidence estimation and calibration methods emerge from the reviewed literature, and what limitations appear when these approaches are applied in multi-agent or pipeline-based LLM systems?
6. **RQ6:** What research gaps and future directions can be identified for confidence calibration and uncertainty quantification in multi-agent LLM systems?

3.2 Study Selection Strategy

The studies included in this review were selected to represent different and complementary stages in the development of confidence and reliability assessment in neural networks, LLMs, and multi-agent LLM systems.

The review uses two related study groups. The first group consists of five foundational studies selected for in-depth conceptual analysis: probabilistic calibration, LLM self-evaluation, consistency-based reasoning, evidence-supported multi-hop reasoning, and broader LLM confidence estimation. The second group consists of six recent studies selected to extend this foundation toward multi-agent confidence calibration and uncertainty quantification.

The five foundational studies are:

1. Guo et al. (**guo2017?**)
2. Kadavath et al. (**kadavath2022?**)
3. Wang et al. (**wang2022?**)
4. Yang et al. (2018) (**yang2018?**)

5. Geng et al. (**geng2024?**)

The recent extension studies examine multi-agent deliberation, uncertainty quantification, runtime calibration, communication-aware calibration, and epistemic calibration in multi-agent planning (**yang2024?**; **liu2025?**; **chen2026matu?**; **armstrong2026margin?**; **huang2026cagecal?**; **wang2026epistemic?**).

The selection is focused rather than comprehensive. The studies are not intended to represent all research on LLM confidence estimation, uncertainty quantification, or multi-agent reliability. Instead, they were selected because together they provide complementary perspectives for tracing the progression from individual-model calibration to system-level multi-agent reliability. The analytical set was selected to maximize conceptual coverage of distinct confidence and reliability signals and assessment levels rather than to maximize the number of included studies. Accordingly, this review is a structured, focused synthesis rather than an exhaustive systematic review or a PRISMA-style evidence map.

The identification process began with the five foundational works and was then extended through a targeted search for multi-agent confidence, uncertainty, calibration, and reliability studies. For the recent extension set, searches were conducted in September 2026 using combinations of terms such as “multi-agent LLM confidence calibration”, “multi-agent uncertainty quantification”, “agent communication calibration”, “runtime confidence calibration”, and “epistemic calibration multi-agent planning” across arXiv, ACL Anthology, OpenReview, and publisher/index pages. Candidate studies were screened for a direct connection to confidence, uncertainty, calibration, or reliability in an interacting-agent setting. The final six extension studies were retained because they collectively cover deliberation, trajectory-level uncertainty, runtime calibration, communication-aware calibration, and planning-level epistemic calibration. This was a targeted and concept-driven search, not an exhaustive database search; therefore, the six recent studies should not be interpreted as a complete census of the multi-agent literature. The final analytical set contains eleven studies.

The selected set also contains both published conference papers and recent preprints. The bibliography uses published versions where available for Wang et al. (**wang2022?**), Geng et al. (**geng2024?**), Liu et al. (**liu2025?**), and Chen et al. (**chen2026matu?**), while Armstrong (**armstrong2026margin?**), Huang et al. (**huang2026cagecal?**), and Wang et al. (**wang2026epistemic?**) are retained as 2026 arXiv preprints.

3.3 Inclusion and Exclusion Criteria

Studies were considered relevant to this review if their main contribution addressed at least one of the following themes:

- confidence calibration or probabilistic reliability;
- confidence or uncertainty estimation in language models;
- model self-evaluation of answer correctness;
- consistency among multiple generated reasoning paths;
- reliability in multi-step or multi-hop reasoning;
- evidence-supported evaluation of generated answers; or
- taxonomies and surveys of confidence estimation and calibration methods for LLMs.

Studies that focused only on improving task performance without a meaningful link to confidence, uncertainty, calibration, or reliability were not included in the core analytical set.

The selection also distinguishes between *confidence methods* and *reliability resources*. A study did not

need to introduce a calibration technique to be included. For example, self-consistency was initially proposed as a reasoning and decoding strategy, not as a calibration method, while HOTPOTQA offers a benchmark for multi-hop reasoning and supporting evidence, not a confidence-estimation algorithm. Their relevance comes from the reliability signals and evaluation angles they provide.

3.4 Categorization Framework

The selected studies are analyzed using two complementary dimensions. The first dimension groups studies according to the primary source of the confidence or reliability signal, while the second dimension identifies the level at which reliability is assessed.

The first dimension groups studies based on the main source of confidence or reliability:

- **Probabilistic calibration:** Confidence comes from the model's predicted probabilities and is checked against actual results.
- **Model self-evaluation:** Confidence is based on how the model judges its own answers or its ability to answer a question.
- **Consistency-based signals:** Reliability is inferred from how consistent or stable the responses or reasoning paths are.
- **Evidence-based evaluation:** Reliability is assessed by examining how correct and relevant the evidence is that supports the final answer.
- **Interaction- and communication-based signals:** Reliability is inferred from agent interactions, communication patterns, disagreement, coordination behavior, or information exchange between agents.
- **Broader confidence-estimation approaches:** These methods use a variety of sources, including model probabilities, internal information, semantic representations, linguistic confidence, consistency, and surrogate models.

The second dimension uses three levels of reliability assessment:

- **Local reliability:** confidence or correctness associated with an individual prediction, answer, or agent output;
- **Process reliability:** reliability of reasoning paths, supporting evidence, communication, or intermediate decisions;
- **System reliability:** reliability of the complete interacting multi-agent system and its final outcome.

This two-dimensional framework is used consistently in the comparative analysis. A study may contribute to more than one level, but the levels are kept conceptually distinct so that evidence about an individual agent is not treated as direct evidence about the complete system.

Analytical Contribution of the Review. The main analytical contribution of this review is the organization of confidence and reliability research into a two-dimensional framework that distinguishes both the source of a reliability signal and the level at which reliability is assessed. The first dimension identifies whether reliability is inferred from probabilistic calibration, model self-evaluation, consistency, evidence, interaction and communication, or broader confidence-estimation signals. The second dimension distinguishes local reliability at the level of an individual agent output, process reliability across reasoning, evidence, communication, and intermediate decisions, and system reliability at the level of the complete interacting multi-agent system.

This distinction provides a common analytical structure for comparing studies that otherwise use different confidence representations, evaluation settings, and objectives. More importantly, it highlights a progression in the reliability problem: confidence that is meaningful at the local level does

not necessarily remain reliable when information is propagated through reasoning steps, transformed through agent interactions, or aggregated into a final system-level decision. Thus, the framework shifts the analysis from asking only how an agent estimates confidence to asking where reliability is measured and how it may change across the local–process–system pipeline.

The framework is therefore used not only to categorize the selected studies, but also to identify where current approaches provide evidence and where connections between reliability levels remain insufficiently addressed. In particular, the reviewed literature provides substantially different forms of evidence at the local, process, and system levels, but does not establish a common mechanism for relating calibrated local confidence to system-level reliability under interaction, dependence, and changing runtime conditions. This observation provides the analytical basis for the research gaps and future directions developed in Sections 8 and 9. The framework is a review-level analytical synthesis of the selected literature rather than a new experimentally validated calibration method.

3.5 Data Extraction and Analysis

For each included study, the following details were collected and examined:

- The study's research goal and the problem it aimed to solve;
- The confidence or reliability signal used;
- The methodology employed;
- The datasets or evaluation tasks involved;
- The metrics and criteria used for evaluation;
- The main findings reported;
- The limitations identified;
- The computational or access requirements relevant to applying the method; and
- The relevance of the study to confidence and reliability in multi-agent systems.

The information was compared qualitatively rather than through numerical analysis because the studies addressed different problems and used different evaluation setups. For instance, temperature scaling is treated as a probabilistic calibration method, self-evaluation examines model knowledge, self-consistency measures agreement among reasoning paths, and HOTPOTQA focuses on multi-hop reasoning and supporting evidence.

The studies are therefore not ranked using a single performance measure. Instead, the analysis considers their underlying assumptions, confidence signals, evaluation strategies, strengths, limitations, and relevance to multi-agent reliability.

Because the literature identification used a targeted, concept-driven search strategy, the choice of search terms, sources, and foundational studies may have influenced the composition of the final analytical set. This limitation is considered when interpreting the synthesis and the research gaps identified from the reviewed literature.

Background

4.1 Confidence Calibration

A model is considered calibrated when the confidence it expresses in its predictions matches the actual probability that those predictions are correct. For instance, if a model gives a confidence rating of 0.8 for a large number of predictions, then about 80% of those predictions should be accurate for the model to be deemed well-calibrated.

Calibration is not the same as predictive accuracy. Accuracy simply measures how often a model makes correct predictions, while calibration assesses whether the confidence in those predictions aligns with the actual likelihood of their correctness. It is possible for a model to have high accuracy but still be poorly calibrated.

4.2 Confidence and Uncertainty

Confidence estimation and uncertainty quantification are related but different ideas. Confidence usually refers to how certain a model is about a particular prediction or answer, whereas uncertainty refers to the lack of certainty in the correctness of the outcome.

In large language model systems, uncertainty can come from various causes, such as a lack of relevant knowledge, ambiguous input, or the existence of multiple possible interpretations or valid answers. These different sources make it challenging to represent uncertainty using just one scalar confidence value.

4.3 Reliability and Trust

In this review, reliability and trust are related but different concepts. Reliability refers to the relationship between a system's outputs, its confidence signals, and the correctness of its predictions under specific conditions. Trust, on the other hand, is a higher-level judgment about whether a system should be trusted in practical use. Trust may be influenced by calibration and reliability, but it is not the same as either: a calibrated confidence score does not automatically imply that a system is trustworthy in all situations.

4.4 Confidence in Multi-Agent LLM Systems

Multi-agent large language model systems add complexity by distributing tasks among multiple interacting agents. In such systems, *local confidence* refers to the reliability estimate for the output of a single agent, while *system-level confidence* refers to the reliability of the final output after the information has gone through several stages of interaction between agents.

For example, a multi-agent system might include agents for tasks like retrieval, reasoning, verification, and response generation. If an intermediate step produces an incorrect result with high confidence, that result could be passed along to later agents, ultimately affecting the final output.This highlights the need to evaluate not only the final output but also the reliability of intermediate steps and how information is shared among agents.

Literature Review

The question of whether a model's stated confidence can be trusted predates the current generation of language models. The selected literature traces a progression from classical neural-network calibration to increasingly complex forms of confidence estimation, reasoning reliability, evidence-based evaluation, and multi-agent system reliability. These studies show how the scope of confidence assessment expands when reliability must be evaluated beyond an individual model output.

5.1 Foundations of Neural Network Calibration

The modern foundation for confidence calibration in neural networks is provided by Guo et al. (**guo2017?**), whose central finding was that improvements in classification accuracy do not necessarily lead to better calibrated confidence estimates. Their experiments examined the relationship between predictive accuracy and calibration and showed that modern neural networks can achieve strong predictive performance while assigning confidence values that do not accurately correspond to observed correctness.

The study investigated several architectural and training factors, including network depth, width, batch normalization, and weight decay, and examined how these factors affected calibration. An important observation was that models could achieve lower classification error while simultaneously exhibiting increasing miscalibration. This demonstrates that accuracy and calibration represent related but distinct properties of a predictive model.

Guo et al. (**guo2017?**) evaluated calibration using reliability diagrams and Expected Calibration Error (ECE). Reliability diagrams provide a visual representation of the relationship between predicted confidence and empirical accuracy, while ECE summarizes the difference between these quantities across confidence intervals.

The study also investigated post-hoc calibration methods, with temperature scaling providing an effective approach. Temperature scaling adjusts the logits of a trained model using a single learned temperature parameter before converting them into probabilities. An important property of this approach is that it can improve the calibration of confidence estimates while preserving the predicted class ordering.

The significance of Guo et al. (**guo2017?**) is therefore twofold. First, the study establishes that predictive accuracy alone is insufficient for evaluating the reliability of confidence estimates. Second, it provides a practical statistical framework for measuring and improving calibration. These concepts form the foundation for subsequent work on confidence estimation in more complex generative models.

5.2 Self-Evaluation in Large Language Models

The transition from conventional classifiers to Large Language Models requires a broader interpretation of confidence because LLMs generate open-ended sequences rather than selecting a single class from a fixed output space. Kadavath et al. (**kadavath2022?**) investigate whether language models can evaluate the correctness of their own answers and estimate whether they know the answer to a question.

The study introduces two important quantities. $P(\text{True})$ represents the model's estimated probability that a proposed answer is correct, while $P(\text{IK})$, where "IK" denotes "I Know", represents an estimate of whether the model can answer a question correctly without depending on a particular proposed answer.

The $P(\text{True})$ formulation extends the idea of confidence beyond conventional class probabilities by asking the language model to estimate the correctness of a specific answer. The results provide evidence that language models can perform useful self-evaluation and that the quality of these estimates can improve with model scale.

The $P(\text{IK})$ formulation addresses a related but different question: whether the model can recognize when it knows the answer to a question. Kadavath et al. (**kadavath2022?**) report that this approach can distinguish questions that models are able to answer from those they cannot answer reliably. However, the calibration of this knowledge estimate can become weaker when evaluated on new task distributions.

The work therefore represents an important transition from probability calibration in conventional classifiers toward self-evaluation in generative language models. The variation in calibration across tasks also highlights the importance of evaluating self-evaluation signals across different task distributions.

5.3 Self-Consistency in Reasoning

A different perspective on reliability is provided by Wang et al. (**wang2022?**), who introduce self-consistency as a decoding strategy for improving chain-of-thought reasoning. Rather than relying on a single reasoning path, the method samples multiple diverse reasoning paths for the same problem and selects the answer that receives the greatest agreement among the sampled solutions.

The primary objective of self-consistency is to improve reasoning accuracy rather than directly calibrate confidence. The method was evaluated across several reasoning benchmarks and produced substantial improvements in task performance. For

example, the reported improvement on GSM8K was 17.9 percentage points, with additional gains reported on SVAMP, AQuA, StrategyQA, and ARC-Challenge.

The relevance of self-consistency to confidence and reliability research comes from its use of agreement as a behavioral signal. Instead of relying exclusively on a model's explicit confidence statement, multiple reasoning paths provide an opportunity to examine whether the model reaches a stable answer under different reasoning trajectories. Importantly, Wang et al. (**wang2022?**) go beyond task accuracy: on GSM8K, they report that the proportion of sampled paths agreeing with the aggregated answer is closely correlated with accuracy (Figure 6), and they suggest using self-consistency as an uncertainty estimate, with low consistency indicating low confidence. They also note that the model's normalized sequence probabilities are not well calibrated, which motivates agreement-based signals as a complementary source of uncertainty information.

However, consistency should not be treated as synonymous with probabilistic calibration. It is a behavioral signal whose usefulness depends on the sampled reasoning paths and the way agreement is constructed. The original study also notes the additional computation required to sample multiple paths. Thus, Wang et al. (**wang2022?**) provide direct evidence that self-consistency can support uncertainty estimation, while leaving open how such a signal should be calibrated and interpreted when reasoning paths are dependent or when multiple agents communicate.

5.4 Multi-Hop Reasoning and Evidence-Based Evaluation

Yang et al. (2018) (**yang2018?**) introduce HOTPOTQA, a dataset designed to evaluate diverse and explainable multi-hop question answering. The dataset contains questions that require models to connect information from multiple supporting documents and provides sentence-level supporting facts associated with the answers.

The multi-hop structure introduces a different dimension of reliability analysis. In a simple prediction task, evaluation may focus primarily on whether the final answer is correct. In multi-hop reasoning, however, an incorrect result may originate from evidence selection, intermediate reasoning, or the connection between multiple pieces of information.

The supporting-fact annotations in HOTPOTQA allow evaluation to consider the evidence used to support an answer in addition to the final answer itself. In the distractor development setting, Yang et al. (2018) (**yang2018?**) report answer EM of 44.44, supporting-fact EM of 21.95, and joint EM of 11.56. The gap between answer correctness and evidence correctness, and the lower joint score requiring both to be correct, illustrates that final-answer accuracy can conceal errors in the supporting reasoning chain. These three measures therefore provide a useful template for multi-agent reporting in which local or intermediate correctness and final system correctness are reported separately and, where appropriate, jointly.

HOTPOTQA does not itself propose a confidence-calibration method. Its relevance is therefore primarily evaluative: it demonstrates why reliability assessment should include intermediate evidence rather than treat final-answer correctness as the only indicator of a successful reasoning process.

5.5 Confidence Estimation and Calibration in LLMs

Geng et al. (**geng2024?**) provide a broader survey of confidence estimation and calibration in Large Language Models. Rather than proposing a single confidence-estimation technique, the survey organizes existing approaches according to the information used to estimate confidence.

The survey considers multiple sources of confidence information, including model probabilities, internal model information, semantic representations, consistency among generated responses, linguistic expressions of confidence, and surrogate-model approaches. It also distinguishes approaches that have access to internal model information from those that rely primarily on observable model behavior. This distinction is relevant to practical LLM applications because closed or API-based systems may expose generated responses without providing access to logits, hidden representations, or other internal states.

Geng et al. (**geng2024?**) further highlight challenges that distinguish LLM confidence estimation from conventional classification calibration. LLMs generate responses from a large output space, multiple responses may express similar semantic meanings, and confidence may be expressed through natural language rather than a numerical probability. A confidence signal must therefore be evaluated not only by whether it can be obtained, but also by whether it corresponds reliably to actual outcomes.

The survey provides a useful framework for interpreting the approaches considered in this review. Probabilistic calibration, model self-evaluation, consistency-based signals, semantic information, and linguistic confidence represent different sources of reliability information. Extending these signals from individual LLM outputs to interacting multi-agent systems introduces additional system-level questions that are examined in the later analysis.

Recent survey work further expands this perspective. Liu et al. (**liu2025?**) provide a comprehensive survey of uncertainty quantification and confidence calibration in LLMs, organizing uncertainty according to dimensions such as input uncertainty, reasoning uncertainty, parameter uncertainty, and prediction uncertainty. The survey also highlights challenges associated with decoding variability, computational requirements, and obtaining reliable uncertainty estimates for generative outputs.

This work extends the broader taxonomy provided by Geng et al. (**geng2024?**) and reinforces the view that LLM uncertainty can arise from multiple sources rather than from a single confidence signal.

5.6 Multi-Agent Confidence Calibration

Recent research extends confidence calibration from individual language models to systems containing multiple interacting agents. The recent studies differ substantially in confidence signal, model access, evaluation data, and maturity of evidence; the results below are therefore reported as author-reported experimental findings rather than as directly comparable benchmarks.

5.6.1 Agreement and Deliberation

Yang et al. (2024) [6] – Collaborative Calibration. The confidence signal is produced through multi-agent deliberation, agreement, and confidence rationalization. The approach is post-hoc and training-free and uses multiple tool-augmented LLM agents, so it mainly requires black-box access to model outputs and the ability to orchestrate additional agent/tool calls. It is evaluated on generative QA tasks across multiple domains and reports improved confidence reliability. The main caveat is that additional calls increase inference complexity, and agreement may not represent independent evidence when agents share models, prompts, or information. The work was published at the ICLR 2024 Workshop on Reliable and Responsible Foundation Models, so it is peer-reviewed workshop evidence rather than a main-track conference result (**yang2024?**).

5.6.2 Trajectory-Level Uncertainty

Chen et al. (2026) [8] – MATU. MATU uses trajectory-level embeddings, communication paths, and communication topology as its uncertainty signal. It requires access to complete multi-agent execution trajectories rather than only final answers, and represents multiple runs as a higher-order tensor before decomposition. The ACL 2026 study evaluates the method across diverse tasks and communication topologies and reports holistic uncertainty estimation. Its caveat is that tensor construction and decomposition introduce representation and computational complexity, and trajectory-level uncertainty is not identical to conventional confidence calibration (**chen2026matu?**).

5.6.3 Runtime Calibration

Armstrong (2026) [9] – MARGIN. MARGIN uses runtime, per-agent calibration factors derived from task-stream observations. It is explicitly black-box: the method requires no model access, held-out calibration set, or retraining. Its evaluation spans 18 foundation models (nine cloud, nine local), 8 benchmarks, and more than 44,000 observations. Under distribution shift, the authors report 3–6× lower calibration error than the strongest design-time baseline. On four of five code-generation benchmarks, raw verbalized confidence reaches only 43.4–50.0% pairwise resolution; calibration raises this to 70.7–86.2% on those benchmarks. The main caveat is operational: online calibration requires maintaining runtime state and observing the task stream. This evidence is currently an arXiv preprint and is therefore less mature than the reviewed conference publications (**armstrong2026margin?**).

5.6.4 Communication-Aware Calibration

Huang et al. (2026) [10] – CAGE-CAL. CAGE-CAL treats communication-induced dependence and counterfactual agreement as the confidence signal. It requires access to agent-level responses and the communication graph so that an observed post-communication graph can be compared with a matched no-communication graph. The study evaluates five benchmarks and reports competitive ECE, improved reliability discrimination, and improved topology selection. The caveat is that counterfactual graph construction and communication modeling add substantial experimental assumptions; the study is also an arXiv preprint rather than a peer-reviewed conference publication (**huang2026cagecal?**).

5.6.5 Epistemic Calibration in Multi-Agent Planning

Wang et al. (2026) [11] – EPC-AW. EPC-AW uses stability of plan evaluation across agents and changing information conditions as an epistemic reliability signal. It requires access to plan evaluations from multiple agents and supports iterative refinement of an epistemic state rather than relying only on final execution success. Experiments report an average 9.75% improvement in system-level success. The caveat is that the evaluation is centered on planning feasibility and epistemic miscalibration, so broader task and architecture generalization remains uncertain; this study is also an arXiv preprint (**wang2026epistemic?**).

Cross-Study Synthesis. Taken together, the reviewed multi-agent studies show an expansion in the reliability question from agreement between agents toward broader system-level behavior. Early approaches examine whether agreement across agents or reasoning paths provides a useful reliability signal. Trajectory-level approaches then consider how uncertainty emerges across sequences of interactions and communication states. Communication-aware approaches further examine how information exchange and dependencies between agents affect agreement and correlated failures, while runtime calibration considers

whether confidence remains reliable as task conditions change during execution. Planning-level approaches extend this perspective by evaluating the stability of epistemic assessments during multi-agent planning. Thus, reliability assessment increasingly moves from asking whether agents agree to asking whether confidence remains meaningful across trajectories, communication, changing runtime conditions, and system-level planning.

Comparative Analysis

The selected studies address confidence and reliability from different methodological perspectives. They are not treated as competing methods under a common benchmark because they differ in objectives, confidence signals, evaluation criteria, task settings, and levels of analysis. The comparison instead asks how each study contributes to the broader problem of reliable confidence assessment as the unit of analysis moves from an individual prediction to a reasoning process and, finally, to an interacting multi-agent system.

The comparison uses the two-dimensional framework introduced in Section 3: six reliability-signal categories and three levels of reliability assessment—local, process, and system. Table 1 summarizes the synthesized answers to the six research questions before the study-level comparison.

Research questions and synthesized findings.

RQ	Synthesized finding
RQ1	Calibration evaluates whether reported confidence corresponds to empirical correctness; ECE and reliability diagrams provide direct probabilistic assessments, while LLM-specific approaches require additional behavioral or semantic signals (**guo2017?**; **geng2024?**).
RQ2	LLMs can provide useful self-evaluations such as P(True) and P(IK), but self-evaluated confidence is not equivalent to conventional probabilistic calibration and can be affected by task and distribution changes (**kadavath2022?**; **geng2024?**).
RQ3	Agreement across sampled reasoning paths provides a useful consistency signal: Wang et al. report that consistency is highly correlated with accuracy and suggest using it as an uncertainty estimate. They also note that the model's sequence probabilities are not well calibrated. Consistency should therefore be treated as a complementary behavioral uncertainty signal rather than as conventional probabilistic calibration (**wang2022?**).
RQ4	Multi-step reliability requires attention to intermediate reasoning and supporting evidence in addition to final-answer correctness; this motivates process-level reliability assessment (**yang2018?**; **chen2026matu?**; **wang2026epistemic?**).
RQ5	The literature can be organized into probabilistic calibration, model self-evaluation, consistency-based signals, evidence-based evaluation, interaction- and communication-based signals, and broader confidence-estimation approaches, operating across local, process, and system levels (**guo2017?**; **kadavath2022?**; **wang2022?**; **yang2018?**; **geng2024?**; **yang2024?**; **liu2025?**; **chen2026matu?**; **armstrong2026margin?**; **huang2026cagecal?**; **wang2026epistemic?**).

RQ	Synthesized finding
RQ6	The multi-agent literature exposes open problems in confidence propagation, correlated errors and disagreement, standardized evaluation, traceability, runtime shift, and computational efficiency (**yang2024?**; **liu2025?**; **chen2026matu?**; **armstrong2026margin?**; **huang2026cagecal?**; **wang2026epistemic?**).

6.1 Study-Level Comparison

The methodological differences among the selected studies are summarized in Table 2. The table records the core method or contribution, primary reliability signal, primary level of analysis, evaluation evidence, and the main methodological limitation or computational consideration reported or implied by the study design. The level shown here represents the study's primary analytical focus; Table 3 provides the more granular mapping of studies across local, process, and system levels.

Comparative analysis of the selected reviewed studies.

Study	Method / Contribution	Reliability Signal	Level	Evaluation Evidence	Main Limitation / Cost
Guo et al. (2017)	Temperature scaling and calibration analysis	Predicted probabilities, ECE, reliability diagrams	Local	Classification calibration; compares confidence with observed correctness	Designed for probabilistic classifiers; does not address generative or multi-agent interaction.
Yang et al. (2018)	HOTPOTQA multi-hop QA benchmark	Supporting facts and final-answer correctness	Process	Answer and supporting-fact evaluation in multi-hop QA	Benchmark/resource rather than a calibration algorithm; reliability depends on evidence annotations.
Kadavath et al. (2022)	Model self-evaluation	P(True), P(IK)	Local	Whether self-evaluated confidence tracks answer correctness and generalizes across tasks	Self-evaluation is distinct from externally verified calibration and can vary with task distribution.
Wang et al. (2023)	Self-consistency through multiple sampled	Agreement among sampled answers	Process	GSM8K, SVAMP, AQuA, StrategyQA, ARC-Challenge; reported gains in reasoning accuracy	Multiple samples increase inference cost; agreement is not itself a calibration guarantee.

Study	Method / Contribution	Reliability Signal	Level	Evaluation Evidence	Main Limitation / Cost
	reasoning paths				
Geng et al. (2024)	Survey and taxonomy of LLM confidence estimation	Probabilities, internal information, semantic, consistency, linguistic, and surrogate signals	Local / Process	Organizes evidence across black-box and white-box approaches	Survey rather than a single evaluated calibration method; signal availability depends on model access.
Yang et al. (2024)	Collaborative Calibration via multi-agent deliberation	Deliberation, agreement, and confidence rationalization	System	Generative QA across multiple domains; evaluates calibration and prediction reliability	Additional agent/tool calls increase inference complexity; shared agents or information can reduce independence of agreement.
Liu et al. (2025)	Survey of LLM uncertainty and calibration	Input, reasoning, parameter, and prediction uncertainty	Local / Process	Synthesizes uncertainty sources, calibration methods, and evaluation challenges	Survey-level evidence; does not itself establish a unified multi-agent benchmark.
Chen et al. (2026)	MATU tensor decomposition of multi-agent trajectories	Trajectory embeddings, communication paths, and topology	Process / System	Experiments across tasks and communication topologies; targets holistic uncertainty estimation	Tensor construction and decomposition add computational and representation complexity.
Armstrong (2026)	MARGIN runtime calibration	Runtime per-agent calibration factors	System	foundation models, 8 benchmarks, and $>$ 44,000 observations; reports 3–6× lower calibration error than the best design-time baseline under	Requires binary correct/incorrect outcome feedback, limiting it to verifiable tasks; maintains online per-agent state.

Study	Method / Contribution	Reliability Signal	Level	Evaluation Evidence	Main Limitation / Cost
				distribution shift	
Huang et al. (2026)	CAGE-CAL counterfactual agent-graph calibration	Communication-induced dependence and counterfactual agreement	System	Five benchmarks; competitive ECE and improved reliability discrimination and topology selection	Requires modeling communication structure and counterfactual graph comparisons.
Wang et al. (2026)	EPC-AW epistemic calibration for multi-agent planning	Stability of plan evaluation across agents and information conditions	Process / System	Reports an average 9.75% improvement in system-level success	Focused on planning feasibility and epistemic miscalibration; broader task generalization remains to be established.

The evaluation measures are not directly interchangeable because they capture different properties. ECE and reliability diagrams directly assess confidence–correctness alignment, whereas $P(\text{True})$ -based evaluation examines self-assessed correctness. Self-consistency primarily measures reasoning agreement and task accuracy, while HOTPOTQA evaluates answer correctness together with supporting evidence. Recent multi-agent work adds trajectory-level uncertainty, communication dependence, runtime adaptation, and planning-level reliability. The comparison therefore emphasizes what each signal can measure and what it cannot establish.

6.2 Reliability-Signal Taxonomy

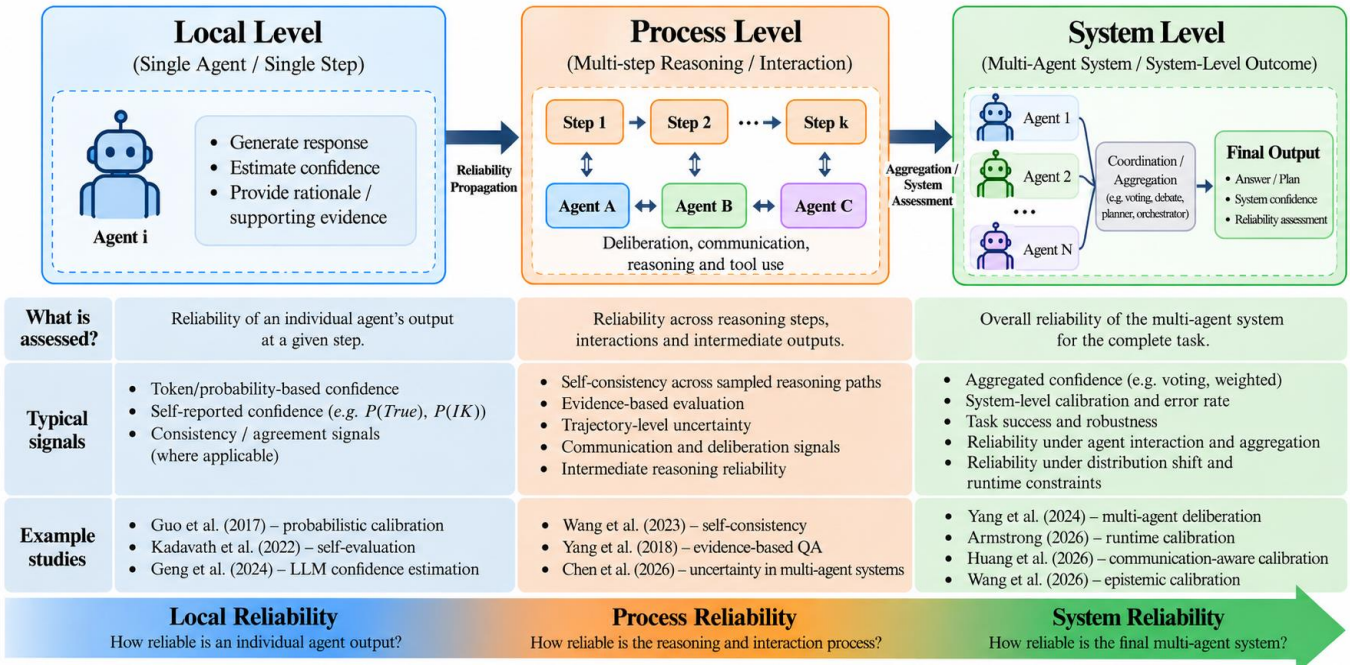
The selected literature is organized using the six reliability-signal categories defined in Section 3. Table 3 presents the same categories as a study-level matrix and maps them to the three common reliability levels. A checkmark indicates that the study substantially contributes to or evaluates the corresponding signal/level combination; it does not imply that the study introduced a complete calibration method at that level.

Reliability-signal and measurement-level taxonomy.

Study	Prob.	Self-eval.	Consistency	Evidence	Interaction	Broader UQ / Confidence	Local	Process	System
Guo et al. (2017)	•						•		
Kadavath et al. (2022)		•					•		
Wang et al. (2023)			•					•	

Study	Prob.	Self-eval.	Consistency	Evidence	Interaction	Broader UQ / Confidence	Local	Process	System
Yang et al. (2018)				•				•	
Geng et al. (2024)	•	•	•			•	•	•	
Yang et al. (2024)		•	•		•				•
Liu et al. (2025)	•	•	•			•	•	•	
Chen et al. (2026)					•	•		•	•
Armstrong (2026)	•				•	•	•		•
Huang et al. (2026)					•	•		•	•
Wang et al. (2026)			•		•	•		•	•

Figure 1 presents a visual representation of the Local–Process–System framework introduced in this review, showing how reliability assessment extends from individual agent outputs to intermediate reasoning and interaction processes and finally to complete multi-agent system outcomes.



Local–Process–System conceptual framework for reliability assessment in multi-agent LLM systems.

The taxonomy and Figure 1 together show that reliability must be considered at different stages of a multi-agent pipeline. The six signal types describe how reliability is estimated, while the local, process, and system levels describe where that reliability is assessed. The levels are therefore complementary rather than mutually exclusive: a single approach may provide evidence at more than one level, while evidence at a lower level does not by itself establish reliability at the system level.

6.3 Relevance to Multi-Agent Reliability

The comparison shows that reliability signals increasingly depend on the unit being evaluated. Foundational studies focus on individual confidence, reasoning consistency, or evidence, whereas recent multi-agent studies add interaction, communication, trajectory, runtime, and planning signals. The taxonomy therefore provides a common vocabulary for the review without assuming that every signal is a calibrated probability.

Wang et al. (**wang2022?**) provide direct evidence that reasoning-path consistency can act as an uncertainty signal, while Yang et al. (2018) (**yang2018?**) show why evidence correctness should be reported alongside final-answer correctness. The recent multi-agent studies then extend the reliability unit to deliberation, complete trajectories, runtime coordination, communication dependence, and planning-level epistemic state (**yang2024?**; **chen2026matu?**; **armstrong2026margin?**; **huang2026cagecal?**; **wang2026epistemic?**). These extensions motivate the research gaps in confidence propagation, dependent agreement, standardized evaluation, traceability, and runtime adaptation developed in the next sections.

This comparison also reveals a central unresolved issue: confidence signals may change in meaning as information moves from individual agents through reasoning and communication processes to system-level aggregation. The reviewed studies address different parts of this pipeline, but no common mechanism establishes how reliability should be propagated and recalibrated across these levels. This observation directly motivates the confidence-propagation and system-level calibration gaps discussed in Section 8.

Cross-Level Synthesis. Across the selected literature, reliability assessment becomes progressively more complex as the unit of analysis moves from an individual output to a reasoning process and then to an interacting multi-agent system. Local methods provide relatively direct confidence–correctness relationships, while process-level approaches introduce reasoning-path consistency, supporting evidence, and trajectory-level uncertainty. Recent multi-agent approaches further introduce communication dependence, runtime adaptation, and system-level aggregation. However, these levels are rarely evaluated within a common framework. In particular, the reviewed studies do not establish how a confidence signal that is reliable for an individual agent should be transformed, propagated, or recalibrated when the corresponding information is communicated to other agents and ultimately incorporated into a system-level decision. This cross-level disconnect represents the central analytical finding of the review and provides a direct basis for the research gaps identified in Section 8.

Critical Discussion

The reviewed literature indicates that confidence can change as information passes through multiple stages of a multi-agent system. This makes reliability, in the sense defined in Section 4.3, an interaction-level concern in addition to an individual-model property; whether a well-calibrated system is also *trustworthy* in deployment is a further judgment this review does not attempt to resolve.

7.1 From Individual Confidence to System-Level Reliability

Calibrated individual agents need not yield a calibrated multi-agent system. This is a system-level inference rather than an experimentally established finding of the foundational studies. Recent multi-agent work begins to address this problem from different perspectives, but the approaches do not yet provide a common mechanism for connecting local confidence to final system-level reliability.

When an output is passed from one agent to another, its confidence may be interpreted, transformed, discounted, or reinforced during subsequent processing. The reliability of the final system may therefore depend on interactions between confidence estimates as well as on the calibration of individual agents.

The reviewed approaches therefore address different failure points rather than competing solutions to a single calibration problem. Agreement-based methods provide a behavioral signal of consistency, trajectory-based methods capture uncertainty across interacting execution paths, runtime calibration addresses changing task conditions, and communication-aware approaches account for dependence introduced by information exchange. These contributions are complementary, but they also reveal that calibration at one level does not automatically provide evidence of reliability at another level. The absence of a common evaluation protocol across these levels remains a central limitation of the current literature.

7.2 Confidence Propagation and Error Accumulation

The extension from individual models to multi-agent systems introduces the problem of confidence propagation. In a multi-agent pipeline, the output of an upstream component can become the evidence or context used by a downstream component. The downstream agent may then make a locally reasonable decision based on information that is already incorrect or incomplete.

Consider a hypothetical pipeline containing retrieval, reasoning, verification, and answer-generation agents. If the retrieval agent selects incomplete evidence, the reasoning agent may nevertheless produce a coherent conclusion. A verification agent may then evaluate the conclusion against the same incomplete evidence and provide additional confidence. In such a situation, multiple stages can reinforce an incorrect intermediate representation.

This example illustrates the distinction between *local correctness* and *global correctness*. An agent may correctly process the information provided to it while the overall system remains incorrect because an earlier stage introduced an error. The confidence of the downstream agent is therefore conditioned on the quality of its input and does not directly represent the confidence of the complete pipeline.

Chen et al. (**chen2026matu?**) further motivate analysis at the level of complete reasoning trajectories by considering uncertainty across cascading reasoning steps and inter-agent communication paths. An important unresolved question is how confidence should be represented, transferred, and recalibrated across a sequence of dependent reasoning stages. Whether errors accumulate, cancel, or become amplified through agent interactions requires dedicated empirical investigation.

7.3 Agent Disagreement and Reliability

Multi-agent systems also introduce a form of uncertainty that is less visible in single-model evaluation: disagreement between agents.

Different agents may reach different conclusions about the same problem because they have different roles, prompts, models, retrieved information, or

reasoning processes. For example, a retrieval agent may identify relevant evidence while a reasoning agent interprets that evidence differently because the two agents perform different sub-tasks. A verification agent may then accept or reject the resulting conclusion.

The meaning of disagreement therefore depends on the roles and information available to the participating agents. Disagreement may reflect genuine uncertainty, differences in task responsibilities, or errors shared across agents.

Huang et al. (**huang2026cagecal?**) further highlight the role of communication structure in correlated failures and false consensus. Their work indicates that agreement between agents can be affected by the way information is exchanged and by the resulting dependencies between agents. Multi-agent calibration should therefore distinguish between independent evidence, role-based disagreement, genuine uncertainty, and disagreement caused by shared or correlated failures.

7.4 Intermediate Reliability and Evidence

Another important issue concerns the level at which reliability should be evaluated. Evaluating only the final response provides information about the outcome of the complete system, but it may conceal the stage at which an error occurred.

Supporting-fact evaluation in HOTPOTQA (Yang et al., 2018) (**yang2018?**) shows that answers can be judged on their evidence, not only their correctness. In the distractor development setting, answer EM is 44.44, supporting-fact EM is 21.95, and joint EM is 11.56. The joint metric is therefore a useful model for multi-agent evaluation: a final answer can be correct while intermediate evidence is wrong, and a joint measure exposes that distinction. The same principle extends to multi-agent systems. A system may require separate evaluation of retrieval quality, intermediate reasoning, verification decisions, and final answer generation. Confidence at each stage may provide information about where uncertainty enters the system and whether downstream confidence reflects the reliability of the information being processed.

This creates a distinction between *outcome-level calibration* and *process-level reliability*. Outcome-level calibration examines whether final confidence corresponds to the correctness of the final result. Process-level evaluation additionally examines whether intermediate decisions, evidence, and planning states are reliable.

Wang et al. (**wang2026epistemic?**) further illustrate the importance of planning-level reliability by examining epistemic calibration during multi-agent planning. These perspectives provide partial foundations for evaluating reliability at different stages, but a unified framework for evaluating confidence throughout a complete multi-agent pipeline is not yet established.

7.5 Evaluation Challenges

Traditional calibration measures such as Expected Calibration Error (ECE) and reliability diagrams provide useful tools for evaluating the relationship between confidence and correctness. However, ECE is a binned approximation, so its value can depend on the number and placement of confidence bins. Guo et al. (**guo2017?**) also show that Maximum Calibration Error (MCE), which reports the largest bin-wise gap, can be particularly sensitive to the binning scheme. Geng et al. (**geng2024?**) further emphasize that LLM confidence is represented through diverse signals and evaluation settings. ECE should therefore not be treated as a complete measure of uncertainty quality.

Confidence may be evaluated independently for each agent, at intermediate reasoning stages, during information exchange, or only at the final system output. These alternatives answer different research

questions and cannot be treated as equivalent. Agent-level calibration can determine whether an individual component provides meaningful confidence estimates; intermediate evaluation can identify stages where confidence becomes misaligned with correctness; and final system-level calibration can assess the complete architecture's output confidence.

A single final calibration metric may therefore conceal miscalibration inside the pipeline, while independent agent-level metrics may fail to capture dependencies between agents. Complementary measures are useful because they answer different questions: AUROC evaluates whether a confidence or uncertainty score discriminates correct from incorrect predictions, whereas AUARC evaluates its usefulness for selective prediction and abstention. Calibration, error discrimination, and selective-prediction utility should therefore be reported together rather than inferred from ECE alone.

Runtime conditions introduce an additional challenge. Armstrong (**armstrong2026margin?**) examines runtime calibration under changing task conditions, highlighting the importance of calibration when system conditions may vary over time. Multi-agent systems may operate under changing task distributions, agent combinations, communication patterns, and evidence conditions. These changing conditions create an additional requirement for calibration methods to remain reliable during system execution.

7.6 Summary

The evaluation discussion shows that multi-agent reliability requires metrics that distinguish calibration from error discrimination and selective-prediction utility, while also identifying where in the pipeline miscalibration occurs.

Research Gaps

The research gaps identified below are derived from the cross-study comparison rather than from the limitations of any single method. They represent recurring discontinuities between the local, process, and system levels of reliability assessment identified in the selected literature.

The gaps below focus on the unresolved problems that become visible when confidence signals are evaluated across interacting agents: connecting local confidence to system outcomes, interpreting dependent agreement, standardizing evaluation, preserving traceability, and maintaining reliability under runtime change.

8.1 Local-to-System Calibration and Confidence Propagation

Existing calibration approaches primarily evaluate the relationship between confidence produced by a model and observed correctness. Guo et al. (**guo2017?**) provide the classical probabilistic foundation, while Kadavath et al. (**kadavath2022?**) examine self-evaluation at the level of individual language-model answers. Recent multi-agent work extends calibration to deliberation, trajectory-level uncertainty, and runtime coordination (**yang2024?**; **chen2026matu?**; **armstrong2026margin?**).

The unresolved problem is how a confidence estimate should change when information passes through dependent reasoning stages. An agent may be well calibrated for its own output while the complete system remains unreliable because an upstream error is propagated, reinforced, or transformed downstream. MATU explicitly motivates trajectory-level analysis of cascading uncertainty (**chen2026matu?**), but existing approaches use different representations and do not establish a common mapping from local confidence to calibrated system-level confidence.

A unified framework should therefore represent, transfer, and recalibrate confidence across dependent agents while separately reporting local, intermediate, and final reliability. The joint answer-and-evidence perspective illustrated by HOTPOTQA (**yang2018?**) provides a useful starting point for such multi-level reporting.

8.2 Correlated Errors and Agent Disagreement

Another important gap concerns the interpretation of agreement and disagreement between agents. Wang et al. (**wang2022?**) demonstrate that agreement among multiple sampled reasoning paths can support reasoning performance. However, multiple reasoning paths generated from the same model may share common assumptions, biases, or failure modes. Similarly, agents in a multi-agent system may share models, training information, retrieved evidence, prompts, or intermediate outputs. Huang et al. (**huang2026cagecal?**) explicitly consider the effects of communication on correlated failures and false consensus. These issues show that agreement cannot be treated as independent confirmation without considering the dependencies between agents and their information sources.

Future calibration methods should account for correlated agent errors while interpreting disagreement according to the roles and information available to participating agents. They should distinguish between independent evidence, role-based disagreement, genuine uncertainty, and disagreement caused by shared or correlated failures.

8.3 Standardized Multi-Agent Calibration Benchmarks

The reviewed literature uses different datasets, tasks, and evaluation settings. Guo et al. (**guo2017?**) focus on conventional classification calibration, Kadavath et al. (**kadavath2022?**) evaluate model self-assessment, Wang et al. (**wang2022?**) evaluate reasoning through multiple sampled paths, and Yang et al. (**yang2018?**) provide a multi-hop reasoning benchmark with supporting evidence. Geng et al. (**geng2024?**) further identify the broader need for comprehensive evaluation of confidence estimation and calibration across tasks and domains.

These resources address different evaluation objectives and do not constitute a standardized benchmark specifically designed to evaluate confidence calibration in multi-agent LLM systems.

A suitable benchmark would need to evaluate not only final-answer correctness but also intermediate agent outputs, confidence estimates, agent disagreement, evidence quality, confidence propagation, and system-level calibration. It should also support comparison across different agent architectures, model combinations, and task types.

The absence of standardized evaluation makes it difficult to compare multi-agent calibration approaches under consistent conditions and to determine whether reported improvements generalize across architectures and task settings.

8.4 Explainable and Traceable Confidence

Confidence estimation is useful not only when a system produces a confidence value, but also when the basis for that confidence can be understood and traced.

Existing confidence-estimation approaches use several sources of reliability information, including probabilities, self-evaluation, consistency, and supporting evidence. Their interaction within a multi-agent system remains insufficiently understood.

In a multi-agent architecture, confidence could be associated with specific evidence retrieved, reasoning steps performed, or verification results. A downstream agent could then evaluate both the information and the confidence associated with that information.

The remaining gap is therefore to make the origin and transformation of confidence traceable across the complete interaction pipeline. Without such traceability, a system-level confidence value may be difficult to interpret or audit when the final decision depends on several intermediate outputs.

8.5 Runtime Shift and Computational Cost

Multi-agent systems may operate under changing task distributions, agent combinations, communication patterns, and evidence conditions. Armstrong (**armstrong2026margin?**) highlights the challenge of maintaining calibration under changing task conditions, while additional sampling, verification, and multi-agent interaction can increase computational cost and latency. Wang et al. (**wang2022?**) demonstrate that generating multiple reasoning paths can improve reasoning performance while requiring additional computation. Future calibration methods should therefore adapt to runtime distribution shift while controlling the computational cost of additional sampling, verification, and agent interaction.

Future Directions

The research gaps identified in the preceding section indicate that future work should move beyond confidence estimation at the level of individual LLM outputs and investigate reliability across complete multi-agent systems. The directions below correspond to the five research gaps identified in Section 8, with each addressing a specific unresolved aspect of multi-agent confidence and reliability.

Research Gap	Future Direction	Evaluation Criteria
Local-to-system confidence propagation	Develop calibration frameworks that track confidence from individual agents through intermediate stages to the final system output.	Evaluate confidence–correctness alignment at agent, intermediate-stage, and system levels using calibration measures such as ECE and Brier Score, including cases where information is transformed across stages.
Correlated errors and agent disagreement	Develop disagreement-aware aggregation methods that account for agent roles, evidence, capabilities, and dependence between agents.	Evaluate whether aggregation distinguishes genuine uncertainty from role-based disagreement and correlated errors while maintaining confidence–correctness alignment.
Lack of standardized multi-agent evaluation	Establish standardized benchmarks covering architectures, model combinations, communication structures, and task types.	Evaluate reproducibility across systems using common measures of local reliability, intermediate-stage reliability, evidence quality, disagreement, confidence propagation, and final system calibration.
Limited explainability of confidence	Develop evidence-linked confidence mechanisms that trace the origin and transformation of confidence across reasoning and interaction stages.	Evaluate traceability of confidence sources, identification of stages where confidence becomes misaligned with correctness, and the reliability of evidence-linked confidence estimates.
Runtime	Develop adaptive calibration methods	Evaluate calibration stability under

distribution shift and computational cost	that respond to changing task conditions while selectively allocating additional sampling, verification, or agent interactions.	distribution shift together with latency, model-call count, computational cost, and the reliability improvement obtained from additional computation.

9.1 Local-to-System Calibration and Confidence Propagation

Future research should develop calibration frameworks that report reliability separately at the agent, intermediate-stage, and final system levels while explicitly representing how confidence is transferred between dependent agents. This direction builds on probabilistic calibration (**guo2017?**), MATU's trajectory-level uncertainty representation (**chen2026matu?**), and the runtime adaptation studied by Armstrong (**armstrong2026margin?**). Success could be evaluated by testing whether propagated confidence remains aligned with correctness after information is transformed through retrieval, reasoning, verification, and evidence-aggregation stages.

9.2 Disagreement-Aware Agent Aggregation

Future research should develop aggregation methods that interpret agent disagreement according to agent roles, available evidence, model capabilities, and the degree of correlation between agents. This direction builds on the agreement-based reliability perspective of Wang et al. (**wang2022?**) and recent communication-aware work on correlated multi-agent failures (**huang2026cagecal?**). Success could be evaluated by testing whether disagreement-aware aggregation improves confidence–correctness alignment while distinguishing genuine uncertainty from role-based disagreement and correlated errors.

9.3 Standardized Multi-Agent Calibration Benchmarks

Future research should establish standardized benchmarks that evaluate multi-agent confidence and reliability across architectures, model combinations, communication structures, and task types. This direction builds on the broad evaluation perspective of Geng et al. (**geng2024?**) and the evidence-aware evaluation structure of HOTPOTQA (**yang2018?**). Success could be evaluated by determining whether the benchmark produces reproducible comparisons across systems and captures both local and system-level reliability.

A standardized benchmark could include the following evaluation dimensions:

- Agent confidence and correctness
- Intermediate-stage reliability
- Evidence quality
- Confidence propagation
- Agent disagreement
- Correlated errors
- Final system calibration

Such a benchmark would allow different multi-agent architectures and confidence-estimation approaches to be evaluated under comparable conditions.

9.4 Explainable and Evidence-Linked Confidence Propagation

Future research should develop confidence mechanisms that link confidence estimates to the evidence, reasoning steps, and verification signals that produced them, while allowing their transformation across agents to be traced. This direction combines the supporting-evidence perspective of HOTPOTQA (**yang2018?**) with the broader confidence sources identified by Geng et al. (**geng2024?**). Success could be evaluated by measuring whether the origin and transformation of a system's confidence can be traced, whether stages

at which confidence becomes misaligned with correctness can be identified, and whether evidence-linked confidence improves reliability assessment.

9.5 Adaptive and Efficient Calibration

Future research should develop calibration methods that adapt to runtime distribution shifts while selectively using additional sampling, verification, or agent interaction when the expected reliability benefit justifies the computational cost. This direction builds on the runtime calibration of Armstrong (**armstrong2026margin?**) and the multi-path reasoning of Wang et al. (**wang2022?**), which illustrates the computational cost of generating additional solutions. Success could be evaluated by measuring calibration stability under changing task conditions together with latency, model-call count, and computational cost relative to the reliability improvement achieved.

Limitations of the Review

This review has several limitations. First, the analysis uses a focused selection of five foundational studies and six recent extension studies rather than a comprehensive literature search. The literature identification followed a targeted, concept-driven search strategy, meaning that the choice of search terms, sources, and foundational studies may have influenced the composition of the final analytical set. Therefore, the findings should not be interpreted as an exhaustive representation of research on LLM confidence, uncertainty quantification, or multi-agent reliability. The relatively small analytical set also limits the breadth with which the identified patterns can be generalized across model families, multi-agent architectures, and application domains.

Second, the reviewed studies differ substantially in their objectives, datasets, confidence representations, evaluation metrics, and experimental settings. As a result, the review uses qualitative comparison rather than quantitative meta-analysis.

Third, the recent multi-agent literature is rapidly evolving and contains heterogeneous approaches to deliberation, uncertainty quantification, runtime calibration, communication-aware calibration, and planning-level epistemic assessment. The synthesis therefore focuses on conceptual relationships among these approaches rather than establishing empirical equivalence between them.

Finally, the unified multi-agent research framework discussed in this review is a review-level synthesis rather than a proposed and experimentally validated calibration method. The conclusions should therefore be interpreted as a structured synthesis and conceptual research direction rather than empirical validation of a new calibration technique.

Conclusion

This review examined the development of confidence calibration and uncertainty quantification from conventional neural-network classification to Large Language Models (LLMs), with particular attention to the challenges that arise when these concepts are extended to multi-agent LLM systems. The reviewed literature shows a progression from individual-model confidence toward broader forms of reliability assessment involving reasoning processes, evidence, agent interaction, communication, and system-level execution.

The foundational literature establishes that predictive accuracy alone is not sufficient to characterize the reliability of confidence estimates. Probabilistic calibration, model self-evaluation, consistency across reasoning paths, evidence-supported evaluation, and broader confidence-estimation approaches provide different ways of assessing reliability. These approaches are

complementary rather than interchangeable because they operate on different confidence signals and evaluate different aspects of model behavior.

Recent multi-agent research extends these perspectives to deliberation, reasoning trajectories, runtime calibration, communication structure, correlated failures, and planning-level epistemic reliability. The comparison indicates that multi-agent systems introduce additional questions that are not captured by evaluating only the confidence of an individual model output. In particular, reliability may depend on how confidence is represented, transferred, transformed, and evaluated as information passes between interacting agents.

The reviewed evidence also suggests that reliability assessment in multi-agent systems should be considered at multiple levels rather than reduced to a single confidence value. Individual agent confidence provides information about local outputs, while intermediate reasoning, evidence, communication, and coordination introduce additional sources of uncertainty that can affect the final system outcome. Consequently, evaluating a multi-agent system requires attention to both the reliability of individual components and the way their confidence and information interact throughout the overall process.

A central gap is therefore the relationship between *local agent confidence* and *system-level reliability*. An individual agent may provide a meaningful confidence estimate for its own output while the complete system remains unreliable because of intermediate evidence, confidence propagation, agent disagreement, correlated errors, communication effects, or changing runtime conditions. Establishing how these factors interact remains an open problem in multi-agent confidence calibration.

The review therefore points to five connected directions: local-to-system calibration and confidence propagation, disagreement-aware aggregation, standardized multi-agent benchmarks, explainable confidence traces, and adaptive calibration under runtime change. These directions should be evaluated with metrics that distinguish calibration, error discrimination, selective-prediction utility, and intermediate evidence correctness.

Overall, confidence calibration and uncertainty quantification remain important components of reliable LLM systems. The main unresolved issue is not the absence of confidence signals, but how heterogeneous signals should be interpreted, propagated, and evaluated when agents interact. A reliable multi-agent framework will therefore need explicit links between confidence, evidence, communication, and final system outcomes.

References

1. C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On Calibration of Modern Neural Networks,” in Proceedings of the 34th International Conference on Machine Learning, pp. 1321–1330, 2017.
2. S. Kadavath et al., “Language Models (Mostly) Know What They Know,” arXiv preprint arXiv:2207.05221, 2022.
3. X. Wang et al., “Self-Consistency Improves Chain of Thought Reasoning in Language Models,” in International Conference on Learning Representations, 2023.
4. Z. Yang et al., “HOTPOTQA: A Dataset for Diverse, Explainable Multi-hop Question Answering,” in Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing, pp. 2369–2380, 2018.

5. J. Geng et al., "A Survey of Confidence Estimation and Calibration in Large Language Models," in Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, pp. 6577–6595, 2024.
6. R. Yang, D. Rajagopal, S. A. Hayati, B. Hu, and D. Kang, "Confidence Calibration and Rationalization for LLMs via Multi-Agent Deliberation," in ICLR 2024 Workshop on Reliable and Responsible Foundation Models, 2024.
7. X. Liu, T. Chen, L. Da, C. Chen, Z. Lin, and H. Wei, "Uncertainty Quantification and Confidence Calibration in Large Language Models: A Survey," in Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining, pp. 6107–6117, 2025.
8. T. Chen, H. Yao, J. Chen, E. E. Papalexakis, and H. Wei, "Every Response Counts: Quantifying Uncertainty of LLM-based Multi-Agent Systems through Tensor Decomposition," in Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics, pp. 16204–16218, 2026.
9. J. Armstrong, "MARGIN: Runtime Confidence Calibration for Multi-Agent Foundation Model Coordination," arXiv preprint arXiv:2605.22949, 2026.
10. J. Huang, M. Li, Z. Li, S. Kwon, H. Yu, and C. Zhang, "Counterfactual Graph for Multi-Agent LLM Calibration," arXiv preprint arXiv:2605.30653, 2026.
11. Z. Wang, S. Jin, Z. Cao, and L. Wang, "When Planning Fails Despite Correct Execution: On Epistemic Calibration for LLM-Based Multi-Agent Systems," arXiv preprint arXiv:2605.23414, 2026.